SENTHIL PUBLIC SCHOOL, DHARMAPURI

AFFILIATED TO CBSE, NEW DELHI (AFFL. NO. 1930248)

CLASS: Class 8 CBSE	**SUBJECT: MATHEMATICS & PHYSICAL SCIENCE**	**DATE: 18-OCT-2026**
TIME: 60 MINS	**PHASE: Phase 1: Sunday Weekly Test**	**MAX. MARKS: 25**

AI GENERATION CONFIRMED: Google Gemini (gemini-3.8-flash) • GROUNDING: Official CBSE Blueprint grounded on Senthil Diary Entries (Oct 12–17)

[Mathematics]

--- SECTION A ---

Q.1 [MTH-P1-Q01] Using the standard algebraic identity $(a - b)^2 = a^2 - 2ab + b^2$, evaluate the expanded expression of $(2x - 3y)^2$: [1] **[1]**
(a) 4x^2 - 9y^2 (b) 4x^2 - 12xy + 9y^2 (c) 4x^2 + 12xy + 9y^2 (d) 4x^2 - 6xy + 9y^2

--- SECTION B ---

Q.2 [MTH-P1-Q02] Solve the linear equation for x by transposing terms systematically and verify your solution: $5x - 7 = 2x + 8$. [2] **[2]**

Q.3 [MTH-P1-Q03] Multiply the binomials $(3a + 4b)$ and $(2a - 5b)$, and simplify the resulting polynomial by grouping and combining all like terms. [2] **[2]**

--- SECTION C ---

Q.4 [MTH-P1-Q04] The area of a trapezium is 168 cm^2 and the perpendicular distance (height) between its parallel sides is 8 cm. If one of the parallel sides measures 18 cm, find the length of the other parallel side. [3] **[3]**

--- SECTION D ---

Q.5 [MTH-P1-Q05] A cylindrical metal water tank open at the top has a base radius of 7 m and a height of 10 m. **[5]**
(a) Calculate the curved surface area (CSA) of the cylinder (use $\pi = \frac{22}{7}$). [2]
(b) Calculate the total surface area of sheet metal required to construct this open-top tank (Base Area + CSA) and the total volume of water it can hold in m^3. [3]

OR

(a) Using standard identities, factorize completely: $49x^2 - 64y^2$. [2]
(b) Solve the following linear equation for x: $\frac{x - 3}{5} + 1 = \frac{2x + 1}{3}$. [3] [5]

[Physical Science]

--- SECTION A ---

Q.6 [SCI-P1-Q06] Ball bearings are widely installed between the hub and axle of bicycles and ceiling fan motors. Which of the following statements correctly explains their functional purpose? [1] **[1]**
(a) They replace sliding friction with rolling friction, substantially decreasing frictional resistance. (b) They increase surface roughness to enhance mechanical grip during high-speed rotation. (c) They eliminate gravitational force acting on the rotating parts. (d) They convert rolling friction into fluid friction to lubricate the joint.

Q.7 [SCI-P1-Q07] Given below are two statements labeled Assertion (A) and Reason (R): **[1]**
Assertion (A): Heavy school bags and backpacks are designed with broad and wide shoulder straps instead of thin cord straps.
Reason (R): For a constant applied load (weight), pressure is inversely proportional to contact area ($P = \frac{F}{A}$); hence, increasing strap surface area reduces pressure on the shoulder.
Select the correct option from the following: [1]
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). (c) Assertion (A) is true but Reason (R) is false. (d) Assertion (A) is false but Reason (R) is true.

--- SECTION B ---

Q.8 [SCI-P1-Q08] Explain why ordinary tap water conducts electricity and causes the magnetic needle of a tester to deflect, whereas pure distilled water does not conduct electricity. State one simple method to convert distilled water into a conducting liquid. [2] **[2]**

--- SECTION C ---

Q.9 [SCI-P1-Q09] A solid rectangular wooden block of mass 6 kg is placed on a flat horizontal table. The base face touching the table has dimensions $0.3\text{ m} \times 0.2\text{ m}$. **[3]**
(a) Taking $g = 10\text{ m/s}^2$, calculate the downward force (thrust) exerted by the block on the table surface. [1]
(b) Calculate the pressure exerted by the block on the tabletop in N/m^2 (Pascals). [2] [3]

--- SECTION D ---

Q.10 [SCI-P1-Q10] (a) Define the process of electroplating. In an electrolytic circuit set up to electroplate copper onto an iron key, identify which electrode (anode or cathode) the iron key should be connected to, and specify the electrolyte solution required. [3] **[5]**
(b) State two key observations demonstrating that liquid pressure increases with increasing depth in a liquid column. [2]

OR

(a) Differentiate between sliding friction and rolling friction with two real-world examples. Explain why rolling friction is always significantly smaller in magnitude than sliding friction. [3]
(b) Why do deep-sea divers wear specially designed reinforced diving suits before venturing deep underwater? State the relationship between liquid pressure, depth, and density. [2] [5]